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ON THE DIFFERENTIABILITY OF THE VALUE FUNCTION IN DYNAMIC MODELS OF ECONOMICS

BY L. M. BENVENISTE AND J. A. SCHEINKMAN¹

WHEN WORKING WITH dynamic models of economic agents, many authors have made use of a differentiability assumption on the valuation function. In the belief that one should understand the implicit restrictions in the model which this assumption imposes, we have found a set of assumptions on the underlying model which guarantee that this condition is met for a certain class of problems.

The class of problems are the infinite horizon calculus of variations problems:

(D) In discrete time, given a technology set $T \subseteq R^{2n}$ find the sequence (x_t) , $t = 0, \dots, \infty$ which solves

$$\text{maximize } \sum_{t=0}^{\infty} u(x_t, x_{t+1}, t)$$

subject to $(x_t, x_{t+1}) \in T$ for all t and x_0 fixed.

(C) In continuous time, given a technology set $T \subseteq R^{2n}$, find the absolutely continuous path $(x(t))$ which solves

$$\text{maximize } \int_0^{\infty} u(x(t), \dot{x}(t), t) dt$$

subject to $(x(t), \dot{x}(t)) \in T$ for all t and $x(0)$ fixed.

Given an initial state x_0 with an associated solution path $y(t, x_0)$ ² the valuation function (see [2]) is defined as:³

$$(D) \quad V(x_0) = \sum_{t=0}^{\infty} u(y(t, x_0), y(t+1, x_0), t),$$

$$(C) \quad V(x_0) = \int_0^{\infty} u(y(t, x_0), \dot{y}(t, x_0), t) dt.$$

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² We assume that the optimal path exists from the point x_0 .

³ The existence of optimal paths is not necessary to define the valuation function. One can define it as

$$V(x) = \sup \int_0^{\infty} u(x, \dot{x}, t) dt$$

over all paths $x(t)$ with $(x(t), \dot{x}(t)) \in T$ for all t and $x(0) = x$.

We will work with a specific initial state x_0 , having the property that the valuation function is defined and finite in a neighborhood of x_0 in R^n , and give conditions which guarantee that V is differentiable at x_0 .

1. A MATHEMATICAL RESULT

The following lemma about concave functions is the main tool used in the paper. The lemma and its proof exist in the literature but we include both for completeness.

LEMMA 1: *Let V be a real valued concave function defined on a convex set $D \subseteq R^n$. If W is a concave differentiable function in a neighborhood N of x_0 in D with the property that $W(x_0) = V(x_0)$ and $W(x) \leq V(x)$ for all x in N , then V is differentiable at x_0 .*

PROOF: We will show that there is only one possible subgradient p for the concave function V at x_0 . Any subgradient p of V at x_0 must satisfy

$$(1) \quad p \cdot (x - x_0) \geq V(x) - V(x_0) \quad \text{for all } x \in D.$$

Since $W(x) \leq V(x)$ in a neighborhood of x_0 and $W(x_0) = V(x_0)$, any vector p satisfying (1) must also satisfy

$$p \cdot (x - x_0) \geq V(x) - V(x_0) \geq W(x) - W(x_0)$$

for all x in the open set N . By the differentiability of W , this p must be unique, and any concave function with a unique subgradient at x_0 , must be differentiable at x_0 (cf. Rockafellar [2, Theorem 25.1, p. 242]). *Q.E.D.*

2. THE DISCRETE TIME CASE

Here we will give sufficient conditions for the differentiability of the value function of problem (D). The following assumptions will be used.

ASSUMPTION 1: T is convex and $\hat{T} \neq \emptyset$.

ASSUMPTION 2: For each t , $u(\cdot, \cdot, t): T \rightarrow R$ is concave and differentiable.

ASSUMPTION 3: An optimal solution (not necessarily unique) $\{y(t, x_0)\}_{t=1}^{\infty}$ exists for (D) at x_0 , and $V(x)$ is well defined for x in some neighborhood of x_0 .

ASSUMPTION 4: $(x_0, y(1, x_0)) \in \hat{T}$.

Assumptions 1 and 2 are the standard convexity assumptions plus a smoothness condition on u , which is clearly needed for the differentiability of V . Assumption 3 requires that the value function be well-defined in a neighborhood of x_0 , and that

an optimal path exists from x_0 , while Assumption 4 requires that the optimal path from x_0 be initially an interior path.

The theorem for discrete time is:

THEOREM 1: *Under Assumptions 1, 2, 3, and 4, V is a differentiable function at x_0 with*

$$V'(x_0) = \frac{\partial}{\partial x} u(x_0, y(1, x_0), 0).$$

PROOF: By the concavity of u and the convexity of T , V is a concave function. Define the function

$$W(x) = u(x, y(1, x_0), 0) + \sum_{t=1}^{\infty} u(y(t, x_0), y(t+1, x_0), t).$$

By Assumptions 3 and 4, W is well defined in an open neighborhood $N(x_0)$ of x_0 . Also since $W(x) = u(x, y(1, x_0), 0) + c$ where c is a constant, by Assumption 2, W is concave and differentiable at x_0 . By construction, $W(x) \leq V(x)$ in $N(x_0)$. Hence, we can apply Lemma 1 to conclude that V is differentiable at x_0 with

$$V'(x_0) = W'(x_0) = \frac{\partial}{\partial x} u(x_0, y(1, x_0), 0). \quad Q.E.D.$$

3. THE CONTINUOUS TIME CASE

In this section, we give sufficient conditions for the value function V associated with problem (C) to be differentiable at a point x_0 .

We retain Assumptions 1 and 3, and give new versions of 2 and 4.

ASSUMPTION 2': $u: T \times R \rightarrow R$ is a continuously differentiable function on $\hat{T} \times R$, and $u(\cdot, \cdot, t): T \rightarrow R$ is concave.

ASSUMPTION 4': An optimal solution $\{y(t, x_0)\}$ exists for the initial state x_0 , and this solution is interior in the following sense: there exists $h > 0$, $\varepsilon > 0$, and $M > 0$ such that for $0 \leq t \leq h$,

$$\|y(t, x_0), \dot{y}(t, x_0)\| < M$$

and if $(z, z') \in R^{2n}$ satisfies

$$\|(y(t, x_0), \dot{y}(t, x_0)) - (z, z')\| \leq \varepsilon \quad \text{for some } t \in [0, h]$$

then $(z, z') \in T$; i.e., there exists an ε -ball around the path contained in T during some initial phase. Also, $V(x)$ is well defined for x in a neighborhood of x_0 .

In order to simplify notation, we will denote by $u_1(u_2)$ the vector of partial derivatives of u with respect to the first n coordinates of R^{2n} (the second n coordinates, respectively). We now state the main theorem of this section.

THEOREM 2: *Under Assumptions 1, 2', 3, and 4', V is continuously differentiable at x_0 with*

$$V'(x_0) = \lim_{h \rightarrow 0} -\frac{1}{h} \int_0^h u_2(y(t, x_0), \dot{y}(t, x_0), t) dt.$$

PROOF: To prove the above, we again will call upon Lemma 1. But in order to do this we will need the following claims:

CLAIM 1: For $\|x - x_0\| \leq \varepsilon(h/(1+h))^{1/2} = \eta$, the path $v(t, x)$ originating at x with

$$\dot{v}(t, x) = \dot{y}(t, x_0) + ((x_0 - x)/h)$$

is a feasible path for $t \in [0, h]$ and $v(h, x) = y(h, x_0)$.

PROOF: By definition

$$\begin{aligned} & \| (y(t, x_0), \dot{y}(t, x_0)) - (v(t, x), \dot{v}(t, x)) \|^2 \\ & \leq \|x - x_0\|^2 + \left\| \frac{x - x_0}{h} \right\|^2 = \left(\frac{1+h}{h} \right) \|x - x_0\|^2 \leq \varepsilon^2. \end{aligned}$$

Hence by Assumption 4', the path $v(x, t) 0 \leq t \leq h$ is feasible. For the second part of the claim, we have

$$\begin{aligned} v(h, x) &= x + \int_0^h \dot{v}(t, x) dt = x + \int_0^h \dot{y}(t, x_0) dt + x_0 - x \\ &= x_0 + \int_0^h \dot{y}(t, x_0) dt = y(h, x_0). \end{aligned}$$

CLAIM 2: The function $w_h: N_\eta(x_0) \rightarrow R$ given by

$$w_h(x) = \int_0^h u(v(t, x), \dot{v}(t, x), t) dt$$

is well defined and concave.

PROOF. To show this, one might observe that if y and $z \in N_\eta(x_0)$, then $v(t, \alpha y + \beta z) = \alpha v(t, y) + \beta v(t, z)$ where $\alpha + \beta = 1$ and likewise for the derivatives. Hence, it is concave because u is concave and well-defined by the previous claim. ■

CLAIM 3: w_h is differentiable at x_0 with

$$\begin{aligned} w'_h(x_0) &= \int_0^h u_1(y(t, x_0), \dot{y}(t, x_0), t) \left[1 - \frac{t}{h} \right] dt \\ &\quad - \int_0^h u_2(y(t, x_0), \dot{y}(t, x_0), t) \frac{1}{h} dt. \end{aligned}$$

PROOF: By the definition of w_h ,

$$(2) \quad w'_h(x_0) = \frac{\partial}{\partial x} \int_0^h u(v(t, x), \dot{v}(t, x), t) dt \quad \text{at } x = x_0.$$

By Assumption 4', the set $\{(v(x, t), \dot{v}(x, t), t) : 0 \leq t \leq h, x \in N_\eta(x_0)\}$ has a compact closure in T . Hence since u is C^1 , we can interchange the differentiation and integration in (2), and obtain

$$\begin{aligned} (3) \quad w'_h(x_0) &= \int_0^h \frac{\partial}{\partial x} (u(v(t, x_0), \dot{v}(t, x_0), t)) dt \\ &= \int_0^h u_1(y(t, x_0), \dot{y}(t, x_0), t) \left[1 - \frac{t}{h}\right] dt \\ &\quad - \int_0^h u_2(y(t, x_0), \dot{y}(t, x_0), t) \frac{1}{h} dt, \end{aligned}$$

since $v(t, x) = y(t, x_0) + x - x_0 + t[(x_0 - x)/h]$. Hence, $w'_h(x_0)$ exists. $\blacksquare$

To complete the proof, we are now in a position to apply Lemma 1. For $x \in N_\eta(x_0)$, define

$$W_h(x) = w_h(x) + \int_h^\infty u(y(x_0, t), \dot{y}(x_0, t), t) dt.$$

Then by definition of V , $W_h \leq V$ in $N_\eta(x_0)$ and $W_h(x_0) = V(x_0)$ with W_h concave and differentiable. Hence, $V'(x_0) = W'_h(x_0) = w'_h(x_0)$. Furthermore, taking limits as $h \rightarrow 0$, we obtain

$$\begin{aligned} V'(x_0) &= \lim_{h \rightarrow 0} \left\{ \int_0^h u_1(y(t, x_0), \dot{y}(t, x_0), t) \left[1 - \frac{t}{h}\right] dt \right. \\ &\quad \left. - \int_0^h u_2(y(t, x_0), \dot{y}(t, x_0), t) \frac{1}{h} dt \right\} \\ &= \lim_{h \rightarrow 0} -\frac{1}{h} \int_0^h u_2(y(t, x_0), \dot{y}(t, x_0), t) dt, \end{aligned}$$

since $\|u_1(y(t, x_0), \dot{y}(t, x_0), t)\| \leq N$, $0 \leq t \leq h$, by Assumptions 2' and 4'. *Q.E.D.*

When the optimal control is piecewise continuous, Assumption 4 will hold for some $h > 0$ provided $(x_0, \dot{y}(0, x_0)) \in \tilde{T}$. Furthermore, in this case we have

$$V'(x_0) = \lim_{h \rightarrow 0} -\frac{1}{h} \int_0^h u_2(y(t, x_0), \dot{y}(t, x_0), t) dt = -u_2(x_0, \dot{y}(0, x_0), 0).$$

Hence we have proved:

COROLLARY 1: *Under Assumptions 1, 2', and 3, if $(x_0, \dot{y}(0, x_0)) \in \tilde{T}$, and if the optimal control is a piecewise continuous function of time, then V is C^1 at x_0 and $V'(x_0) = -u_2(x_0, \dot{y}(0, x_0), 0)$.*

To conclude, the conditions required for the result seem to be the minimum required for the general result. It might also be pointed out that these results give a

good economic characterization of the shadow prices that arise from the calculus of variations. It follows from Corollary 1, that the shadow prices from the calculus of variations represent marginal valuations of capital. This result has implications about the necessity of the transversality condition for such problems. This is carried out in a more general framework in Benveniste and Scheinkman [1].

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